VIT-AP UNIVERSITY · B.TECH | CSE (AI & ML)

SITEWISE AI

Geospatial intelligence for smarter land-use decisions

AI-assisted satellite-image land-cover analysis and project-defined site-suitability screening.

VIVA · RESEARCH DEMO

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01 — THE SCREENING PROBLEM

From satellite images to intelligent land-cover insights

Manual interpretation, inconsistent classification, image corruption, and land-cover-only reasoning need one workflow for prediction, reliability, and ranked screening.



INPUT

Satellite image

A compact RGB scene carries the evidence.



PREDICT

Land-cover probabilities

The classifier estimates ten possible classes.



SCREEN

Ranked site-use suggestions

Project-defined categories are ordered with evidence quality in view.

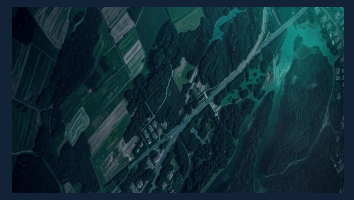
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02 — DATA FOUNDATION

Built on Sentinel-2 satellite imagery



FOREST



AGRICULTURE



RESIDENTIAL

Source	Image format	Dataset entries	Evaluation framing
EuroSAT RGB from tanganke/eurosat	64×64 RGB resized to 224×224×3 in preprocessing	21,600 training 2,700 clean-test 7 × 2,700 corrupted splits	10 classes Split entries are not unique images

CLASS ORDER

AnnualCrop · Forest · HerbaceousVegetation · Highway · Industrial · Pasture · PermanentCrop · Residential · River · SeaLake

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03 — MODEL ARCHITECTURE

EfficientNetB0 — The core classification model

01	02	03	04
Input	**Preprocess**	**Backbone**	**Head**
64×64 RGB	Resize to 224×224×3	EfficientNetB0	Global average pooling → dense layer

OUTPUT

Softmax

Ten class probabilities form the land-cover prediction.

TRAINING STRATEGY

- Frozen-backbone training
- Staged fine-tuning
- BatchNormalization remains frozen
- Augmentation supports robustness

Verified path: RGB input → spatial features → pooled representation → class probabilities. No layer sizes or dropout rates are assumed.

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04 — INFERENCE LOGIC

How the AI makes a land-cover prediction

1

Image

One RGB scene enters the model.

2

Logits

The head produces one score per class.

3

Softmax

Scores normalize into ten probabilities.

4

Prediction

The largest probability becomes the leading class.

ILLUSTRATIVE EXAMPLE

Forest · 72%

The model's highest softmax output is Forest.

NORMALIZATION

$$p_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

Softmax confidence is not automatically calibrated correctness probability.

TREAT THIS AS MODEL OUTPUT; VALIDATE CALIBRATION AND CORRECTNESS SEPARATELY.

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05 — RELIABILITY + EVIDENCE

Reliability screening makes the prediction inspectable



Clean input

The land-cover classifier sees usable evidence.



Degraded input

A separate reliability CNN screens evidence quality.

PROJECT THRESHOLDS

≥ 0.70

Land-cover confidence

≥ 0.55

Reliability score

EIGHT DETERMINISTIC RGB-DERIVED FEATURES

Color and texture cues support evidence-quality adjustment: mean and spread of RGB channels, channel relationships, brightness, saturation, and related deterministic image statistics.

Boundary: these features are not segmentation and do not identify object boundaries or land parcels.

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06 — SCREENING LOGIC

From classification to ranked site-use suggestions

01

10 probabilities

Land-cover distribution from EfficientNetB0.

02

8 visual features

Deterministic RGB-derived evidence cues.

03

18D representation

A combined input for project-defined screening.

04

Ranked suggestions

Eight recommendation categories are ordered.

WHAT THE SCORING LAYER DOES

Applies project-defined scoring rules, adjusts for evidence quality, and returns a ranked screening view.

SCIENTIFIC BOUNDARY

Scores are not probabilities.

Suggestions are preliminary screening outputs—not construction approval or validated real-world site-selection prediction.

THE SYSTEM RANKS PROJECT-DEFINED OPTIONS; IT DOES NOT REPLACE FIELD VERIFICATION, PLANNING REVIEW, OR DOMAIN JUDGMENT.

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07 — APPLICATION FLOW

An end-to-end AI application

01

01

Browser

User submits an image and requests analysis.

02

02

Flask

Application routes receive and validate the request.

03

03

Models

Classifier and reliability CNN process the image.

04

04

JSON

Predictions, evidence, and rankings return as structured output.

05

05

Interface

Results become readable land-cover insights.

TECHNOLOGY STACK

Browser · Flask · EfficientNetB0 · Reliability CNN · JSON · Render

Render is the intended deployment platform, not a claim of live deployment.

ROUTE CONTRACT

Request → inference → JSON response

The application boundary keeps model output structured and explainable to the front end.

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08 — EVALUATION + LIMITATIONS

Evaluating reliability and recognizing limitations

PLANNED EVALUATION SUITE

Measure	Status	Interpretation
Accuracy	Pending	Overall classification
Precision / Recall	Pending	Class-level behavior
F1 Score	Pending	Balance of precision and recall
mIoU	Pending	Use only where the evaluation definition supports it

LIMITATION 01 · Cloud cover — Limits visibility and prediction accuracy.

LIMITATION 02 · Similar land covers — Some classes are visually similar.

LIMITATION 03 · Resolution constraints — Small or mixed objects may be missed.

HISTORICAL PROJECT-ARTIFACT METRICS

0.000075

MSE

0.004929

MAE

98.67%

Top-1 recommendation agreement

Historical artifact metrics only.

FUTURE SCOPE: STRONGER CALIBRATION, BROADER GEOGRAPHIC VALIDATION, RICHER EVIDENCE, AND FIELD-LEVEL VERIFICATION.

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SITEWISE AI
Geospatial Intelligence for Smarter Land Use Decisions

CLOSING FRAME

From satellite imagery to explainable land-cover insights

An integrated workflow connects image evidence, model output, reliability screening, and project-defined site-use suggestions.

SEE
Classify the land cover.

CHECK
Screen reliability and evidence quality.

SCREEN
Rank project-defined site-use suggestions.

Suitability claims remain preliminary screening outputs, not construction approval or validated real-world site-selection prediction.

THANK YOU · QUESTIONS & ANSWERS

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